

Article

RWKV-CVM: Gated Cross-Variate Mixing for Multivariate Power Load Forecasting

Adil Rizki , Abdelwahed Echchatbi and Hamid Yantour 

Laboratory of Management, Innovation and Engineering (LMII)—Faculty of Sciences and Technology,
Hassan 1st University, Settat 26000, Morocco

* Correspondence: adil.rizki@uhp.ac.ma

Abstract

Accurate power load forecasting is essential for efficient electricity grid management, yet capturing cross-variate dependencies in multivariate time series remains a persistent challenge. Recent channel-independent methods based on Transformer and recurrent architectures have achieved strong forecasting performance, but they discard potentially useful information from correlated variates such as weather conditions and neighboring consumption zones. In this paper, we propose RWKV-CVM, a lightweight extension of the RWKV-TS architecture that introduces a trainable Cross-Variate Mixing (CVM) module to selectively incorporate inter-variate information while preserving the linear time complexity of the backbone. The CVM module is a gated, row-stochastic mixing matrix—initialized from the training set absolute Pearson correlations and modulated by a single learned scalar gate that is applied to the normalized input series before patching, adding only 65 trainable parameters to the backbone. We evaluate the method under a single unified harness (three random seeds, consistent normalization, and re-executed DLinear, iTransformer and RWKV-TS baselines) on three settings: the Tetouan city power consumption dataset forecast jointly for all three zones at horizons up to 72 h (including the operationally relevant 24 h day-ahead and 48 h two-day-ahead horizons) and the ETTh1 and Weather benchmarks under a 10% few-shot protocol. Averaged over horizons, RWKV-CVM attains the lowest mean MSE on all three datasets (Tetouan all-zone 0.0427, ETTh1 0.640, Weather 0.250), narrowly ahead of the strongly-tuned baselines and its own RWKV-TS backbone. The advantage is modest, is concentrated at longer horizons, and is selective across target zones; on several individual horizons and in the full-data regime, a baseline is preferable, and we report these cases explicitly. These results indicate that a controlled, lightweight injection of cross-variate information can improve multivariate load forecasting on average without sacrificing computational efficiency.

Keywords: power load forecasting; RWKV; cross-variate mixing; multivariate time series; deep learning; channel-independent modelling; few-shot forecasting



Academic Editor: Pavlos Georgilakis

Received: 29 March 2026

Revised: 7 June 2026

Accepted: 15 June 2026

Published: 18 June 2026

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1. Introduction

1.1. Background and Literature Review

In 2021, the share of electricity in global final energy consumption reached 20.4% [1]. Electricity consumption has been increasing steadily worldwide, making accurate power load forecasting an indispensable component of modern grid management. If there is no accurate prediction of power load, it will lead to either excessive or insufficient power

generation facilities, both of which have serious economic and social consequences [2]. Reliable forecasts enable power utilities to optimize generation scheduling, reduce operational costs, and maintain the dynamic balance between supply and demand [3]. Forecasting accuracy is particularly critical at the distribution level, where consumption patterns are influenced by a complex interplay of meteorological factors, temporal periodicity, and spatial dependencies among neighboring zones [4,5].

From a methodological perspective, traditional modeling approaches based on statistical analysis have good interpretability. The autoregressive moving average model (ARMA) is a classical method that has been applied to short-term power forecasting [6]. However, ARMA is only suitable for stationary processes. The autoregressive integrated moving average (ARIMA) model extends ARMA by converting non-stationary sequences into stationary ones through differencing [7], but it still cannot effectively handle nonlinear patterns or multivariate time series.

Artificial intelligence (AI) modeling is data-driven and has been widely applied to power load forecasting. Backpropagation (BP) neural networks [8] and artificial neural networks (ANNs) [9] were among the early deep learning approaches. Recurrent neural networks (RNNs) improved upon these by retaining temporal information through self-connected structures [10], but they suffer from vanishing/exploding gradient problems and cannot effectively capture long-term dependencies. The LSTM model [11] alleviates this problem through gating mechanisms, making it suitable for both short-term and long-term power prediction [12]. Kim and Cho [13] further proposed a CNN-LSTM hybrid that combines convolutional feature extraction with recurrent modeling.

The Transformer architecture [14] has been increasingly adopted for time series tasks. Its self-attention mechanism enables data at each time point to attend to all other time points, solving the long-term dependence problem that plagues RNNs. Several variants have been proposed to address the quadratic complexity of the vanilla Transformer. The Informer [15] replaces full attention with ProbSparse self-attention and introduces a distilling mechanism. The Autoformer [16] incorporates seasonal decomposition and an auto-correlation mechanism. FEDformer [17] operates in the frequency domain using a mixture of experts. The Reformer [18] employs locality-sensitive hashing to reduce complexity to $O(L \log L)$.

Beyond Transformers, CNN-based models such as TimesNet [19] model temporal 2D-variations for general time series analysis, achieving state-of-the-art results across multiple benchmarks. On the MLP front, DLinear [20]—a simple linear model—demonstrated that many Transformer-based approaches do not significantly outperform linear baselines, sparking renewed interest in model efficiency. This led to the development of numerous lightweight MLP models, including TSMixer [21] and TiDE [22]. Meanwhile, PatchTST [23] demonstrated that segmenting univariate time series into patches and processing them with a standard Transformer yields state-of-the-art performance while significantly reducing computational cost.

In the power domain specifically, Wang et al. [3] introduced an ensemble learning framework combining LSTM and the Informer model for the Tetouan city power consumption dataset. Their LSTM-Informer architecture uses LSTM as a base learner to capture short-term correlations while the Informer handles long-term dependencies, achieving the best reported results among six baselines on this dataset. However, their approach relies on a two-layer ensemble that increases architectural complexity and training overhead.

More recently, the RWKV-TS model [24] represents a promising alternative that adapts the Receptance Weighted Key Value (RWKV) architecture—originally designed for language modeling by Peng et al. [25]—to time series tasks. RWKV-TS achieves $O(L)$ time and memory complexity through a linear RNN formulation while maintaining competitive accuracy. Compared to TimesNet, RWKV-TS demonstrates a relative 12.58% average MSE

reduction for long-term forecasting while requiring significantly less training time (0.067 s vs. 5.723 s per step) [24]. However, RWKV-TS follows a strictly channel-independent strategy, processing each variate separately.

A separate but equally important debate concerns the channel-dependent (CD) versus channel-independent (CI) paradigm in multivariate time series (MTS) forecasting. CD methods jointly model multiple variates through architectures such as graph neural networks (MTGNN [26]), cross-attention Transformers (Crossformer [27], iTransformer [28]), and mixer layers. However, CI methods that process each variate independently have often outperformed CD approaches [23,29], largely because CD methods tend to overfit spurious correlations, especially when inter-variate relationships are non-stationary.

Zhao and Shen [30] offered a nuanced perspective by highlighting the locally stationary lead-lag relationships between variates. Their LIFT method dynamically identifies leading indicators and leverages them in the frequency domain, achieving an average improvement of 5.4% over state-of-the-art CI methods across six benchmark datasets. This suggests that the optimal approach lies between the two extremes: selectively incorporating cross-variate information when it is reliable, while avoiding indiscriminate mixing that introduces noise.

1.2. Research Gap

Through the literature review, we identify the following gaps:

- While the RWKV-TS model achieves strong temporal modeling with $O(L)$ complexity, it follows a strictly channel-independent strategy, discarding potentially useful cross-variate information such as the well-documented correlations between neighboring consumption zones ($\rho = 0.75$ between Zone 1 and Zone 3) and weather variables ($\rho = 0.49$ for temperature).
- Existing CD methods either mix all variates indiscriminately (introducing noise) or employ complex architectures (e.g., graph neural networks [26], cross-attention mechanisms [27], or frequency-domain mixers [30]) that substantially increase model complexity. There is no lightweight mechanism to selectively inject cross-variate information into an efficient linear-RNN backbone.
- On the Tetouan power consumption dataset specifically, the best reported results (by Wang et al. [3]) rely on a two-layer LSTM-Informer ensemble, leaving room for architectures that can better exploit the strong inter-zone correlations and lead-lag dynamics documented in the data.
- Previous studies on this dataset [3,31] have not considered cross-correlation analysis or lead-lag stationarity assessment, limiting the understanding of temporal inter-variate dynamics that could inform model design.

1.3. Contribution

To address these gaps, this paper proposes **RWKV-CVM** (RWKV with Cross-Variate Mixing), a lightweight extension of RWKV-TS that introduces a learnable mixing module to selectively share information between variates. The CVM module computes a gated linear combination of the (instance-normalized) input variates, controlled by a row-stochastic mixing weight matrix $\mathbf{W}$ and a single scalar gate α and is applied *before* patching; each variate is thereafter processed independently by the channel-independent RWKV-TS backbone. We use a single, data-driven correlation-based initialization for $\mathbf{W}$, derived from the absolute Pearson correlation coefficients computed on the training data. The proposed approach is evaluated under one unified experimental harness—three random seeds (42/43/44), consistent normalization across all models, and re-executed DLinear, iTransformer and RWKV-TS baselines—on the Tetouan dataset forecast jointly for all three zones (horizons {24, 48, 144, 288, 432}, steps, i.e. 4, 8, 24, 48 and 72 h) and, to assess generaliza-

tion, on the ETTh1 and Weather benchmarks under a 10% few-shot protocol (horizons $\{96, 192, 336, 720\}$).

The main contributions of this paper are as follows:

- The Pearson correlation coefficient, Random Forest feature importance, and cross-correlation function (CCF) were used to perform a comprehensive analysis of the inter-variate dependencies in the Tetouan power dataset. This analysis reveals strong lead-lag relationships between zones (Zone 1 leads Zone 3 by ~ 20 min with $|R| = 0.755$) and non-trivial weather–consumption dynamics (temperature leads Zone 3 by ~ 4.7 h), providing an empirical foundation for cross-variate modeling. We additionally report an Augmented Dickey–Fuller (ADF) stationarity test for all channels and a train-versus-evaluation correlation-drift check that confirms the correlation prior is computed without test-set leakage.
- We propose the CVM module, a plug-and-play component that augments channel-independent time series models with controlled cross-variate information sharing through a gated mixing mechanism ($\tilde{\mathbf{X}} = (1 - \alpha)\mathbf{X} + \alpha\mathbf{X}\mathbf{W}^\top$). The module adds only 65 trainable parameters ($M^2 + 1$) to the backbone, preserving $O(L)$ computational complexity. The mixing matrix is initialized from the training-set absolute correlations, which provides a data-driven prior on which variates are likely to share useful information.
- We evaluate RWKV-CVM against re-executed DLinear, iTransformer and RWKV-TS baselines under one consistent normalization and three-seed harness, across the Tetouan all-zone (joint) setting—now including the operationally relevant day-ahead (24 h) and two-day-ahead (48 h) horizons—and the ETTh1 and Weather few-shot benchmarks.
- Averaged over horizons, RWKV-CVM attains the lowest mean MSE on all three datasets, narrowly ahead of the strongly tuned baselines and its RWKV-TS backbone. The improvement is small, is concentrated at longer horizons, and is selective across target zones (it helps Zone 1 and Zone 2 but not the more autonomous Zone 3). We do not claim uniform or full-data superiority, and we report the individual horizons and the full-data regime in which a baseline is preferable.

The remainder of this paper is organized as follows. Section 2 formulates the forecasting problem mathematically. Section 3 presents the proposed RWKV-CVM architecture. Section 4 describes the dataset, exploratory analysis, experimental configuration, and evaluation metrics. Section 5 reports and discusses the results, including the joint all-zones comparison, the few-shot generalization study, per-zone transferability, the learned gate and mixing structure, complexity, and a discussion of why the gated mixing helps. Finally, Section 6 concludes the paper.

2. Problem Statement

In this work, we assume that we can obtain the weather data $\vec{W}$ of the power grid during the observation period and the corresponding power load data $\vec{P}$ for three distribution zones. The available weather data include temperature ($\vec{Temp}$), humidity ($\vec{Hum}$), wind speed ($\vec{Ws}$), and solar irradiance measures (GeneralDiffuseFlows, DiffuseFlows).

Formally, the power load at time step t on day d is denoted $P(t, d)$. The multivariate input vector at each time step is

$$\vec{x}(t, d) = [P_1(t, d), P_2(t, d), P_3(t, d), \vec{W}(t, d)] \quad (1)$$

where P_1, P_2, P_3 are the power consumption of Zones 1, 2, and 3, respectively, and $\vec{W}(t, d) = [\text{Temp}(t, d), \text{Hum}(t, d), \text{Ws}(t, d), \text{GDF}(t, d), \text{DF}(t, d)]$ contains the weather features.

Given a multivariate time series $\mathbf{X} \in \mathbb{R}^{L \times M}$ consisting of L historical time steps and $M = 8$ variates, the forecasting task is to predict the future H time steps:

$$\hat{\mathbf{Y}} = f_{\theta}(\mathbf{X}) \in \mathbb{R}^{H \times M} \quad (2)$$

where f_{θ} denotes the forecasting model with parameters θ . In contrast to the original single-target formulation, we forecast all three zones jointly and optimize a multivariate objective over all forecast channels (Section 3) so that the value of cross-variate sharing can be assessed consistently across target variables. We use $L = 288$ steps (48 h) as the input length and evaluate forecasting performance at five horizons $H \in \{24, 48, 144, 288, 432\}$ steps, corresponding to 4, 8, 24 (day-ahead); 48 (two-day-ahead); and 72 h ahead. The 24 h and 48 h horizons are the operationally relevant generation-scheduling windows motivated in Section 1; they replace the 12/16/20 h horizons of the original submission, which omitted the day-ahead range.

The key challenge addressed in this work is understanding how to effectively exploit the inter-variate dependencies—particularly the strong correlations between the consumption zones and weather variables—while avoiding the overfitting issues that plague channel-dependent methods. This motivates the design of the Cross-Variate Mixing module presented in the next section.

3. Methodology

This section presents the proposed RWKV-CVM architecture. We first describe the RWKV-TS backbone, then detail the Cross-Variate Mixing (CVM) module that extends it.

3.1. RWKV-TS Backbone

The RWKV-TS backbone processes each variate $\mathbf{x}_i \in \mathbb{R}^L$ independently through the following pipeline (see Algorithm 1 for an overview). Its internal time-mixing and channel-mixing blocks are shown in Figure 1.

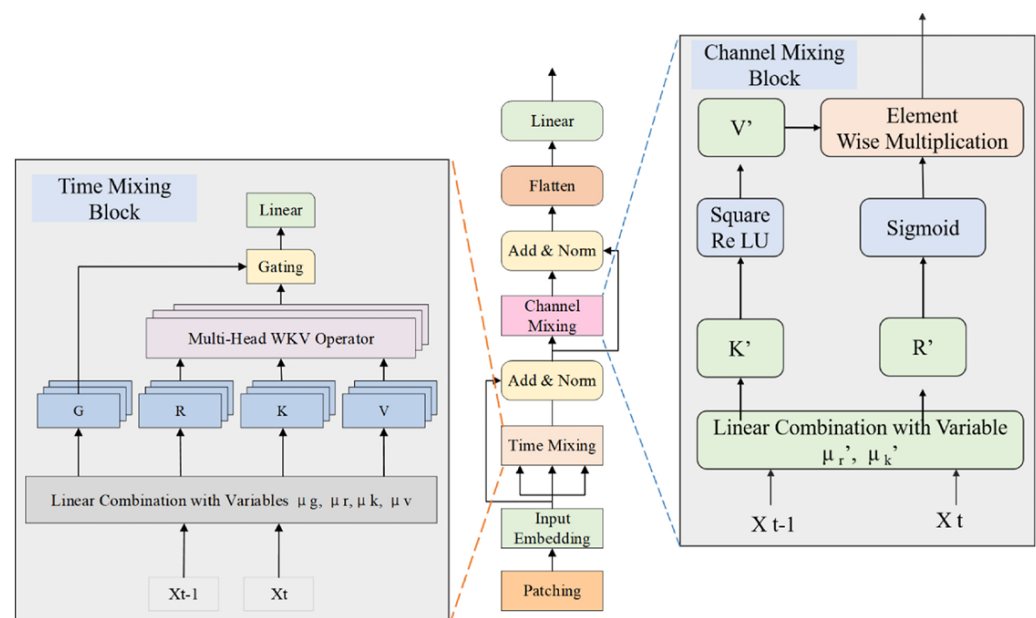


Figure 1. The architecture of the RWKV-TS backbone, showing the central flow (patching, input embedding, stacked time-mixing and channel-mixing blocks, flatten and linear head) together with the internal structure of the time-mixing block (left) and the channel-mixing block (right). Each variate is processed independently along this path.

Algorithm 1 RWKV-CVM Forward Pass

Require: Input $\mathbf{X} \in \mathbb{R}^{L \times M}$, prediction horizon H
Ensure: Forecast $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times M}$

- 1: **for** each variate $i = 1, \dots, M$ **do**
- 2: $\bar{\mathbf{x}}_i \leftarrow \text{InstanceNorm}(\mathbf{x}_i)$ ▷ Normalize per variate
- 3: **end for**
- 4: $\bar{\mathbf{X}} \leftarrow [\bar{\mathbf{x}}_1, \dots, \bar{\mathbf{x}}_M]$ ▷ Stack normalized variates
- 5: **if** CVM enabled **then**
- 6: $\mathbf{W} \leftarrow \text{Softmax}_{\text{row}}(\mathbf{W}_{\text{raw}})$ ▷ Row-stochastic mixing matrix
- 7: $\alpha \leftarrow \sigma(a_{\text{raw}})$ ▷ Sigmoid gate
- 8: $\tilde{\mathbf{X}} \leftarrow (1 - \alpha)\bar{\mathbf{X}} + \alpha \bar{\mathbf{X}}\mathbf{W}^T$ ▷ Cross-variate mixing (before patching)
- 9: **else**
- 10: $\tilde{\mathbf{X}} \leftarrow \bar{\mathbf{X}}$ ▷ Channel-independent
- 11: **end if**
- 12: **for** each variate $i = 1, \dots, M$ **do**
- 13: $\mathbf{x}_i^p \leftarrow \text{Patch}(\tilde{\mathbf{x}}_i; P, S)$ ▷ Create patches
- 14: $\mathbf{x}_i^t \leftarrow \text{LinearProjection}(\mathbf{x}_i^p)$ ▷ Project to D
- 15: **for** each layer $\ell = 1, \dots, N_{\text{layer}}$ **do**
- 16: $\mathbf{x}_i^t \leftarrow \mathbf{x}_i^t + \text{TimeMixing}_{\ell}(\mathbf{x}_i^t)$ ▷ WKV attention
- 17: $\mathbf{x}_i^t \leftarrow \mathbf{x}_i^t + \text{ChannelMixing}_{\ell}(\mathbf{x}_i^t)$ ▷ FFN
- 18: **end for**
- 19: $\mathbf{h}_i \leftarrow \text{Flatten}(\mathbf{x}_i^t)$ ▷ Flatten patches
- 20: $\hat{\mathbf{y}}_i \leftarrow \text{InverseNorm}(\text{LinearHead}(\mathbf{h}_i))$ ▷ Project to H , denormalize
- 21: **end for**
- 22: **return** $\hat{\mathbf{Y}} = [\hat{\mathbf{y}}_1, \dots, \hat{\mathbf{y}}_M]$

Instance Normalization. Each univariate time series is normalized to zero mean and unit standard deviation using its own statistics. The normalization parameters are stored and reapplied to the output predictions through a reverse normalization step [32].

Patching. The normalized series is segmented into N overlapping or non-overlapping patches of length P with stride S . The number of patches is computed as

$$N = \left\lfloor \frac{L - P}{S} \right\rfloor + 2 \quad (3)$$

Each patch $\mathbf{x}_i^p \in \mathbb{R}^{N \times P}$ is projected to the model dimension D through a learnable linear layer, yielding input tokens $\mathbf{x}_i^t \in \mathbb{R}^{N \times D}$.

RWKV Encoder. The encoder consists of N_{layer} stacked residual blocks. Each block contains two sub-blocks:

- A *time-mixing* sub-block that implements the WKV attention mechanism with token shifting and exponential time decay, enabling efficient sequential information aggregation.
- A *channel-mixing* sub-block that applies a gated non-linear transformation using the squared ReLU activation function.

The time-mixing sub-block interpolates the current token x_t with the previous token x_{t-1} through learnable mixing coefficients $\mu_g, \mu_r, \mu_k, \mu_v$, and then computes the multi-head WKV attention in parallel mode. In the recurrent mode, the WKV operator can be expressed as

$$\text{wkv}_t = s_{t-1} + \text{diag}(\mathbf{u}) \cdot \mathbf{k}_t^T \cdot \mathbf{v}_t \quad (4)$$

$$s_t = \text{diag}(\mathbf{w}) \cdot s_{t-1} + \mathbf{k}_t^T \cdot \mathbf{v}_t \quad (5)$$

where s_t is the hidden state at time step t , and $\mathbf{w}$ and $\mathbf{u}$ are learned decay and bonus parameters, respectively.

Flatten and Linear Head. The output of the last RWKV block is flattened across the patch dimension and projected to the prediction length H through a linear layer, producing the per-variate forecast $\hat{\mathbf{y}}_i \in \mathbb{R}^H$.

3.2. Cross-Variate Mixing (CVM) Module

The key contribution of this work is the CVM module, which introduces controlled cross-variate information sharing. In contrast to the original submission—where the mixing was described as acting on the encoder outputs—the module is applied to the instance-normalized input series *before* patching, so that the cross-variate correction propagates through the entire channel-independent backbone. This placement matches the implementation and is reflected in Figure 2.

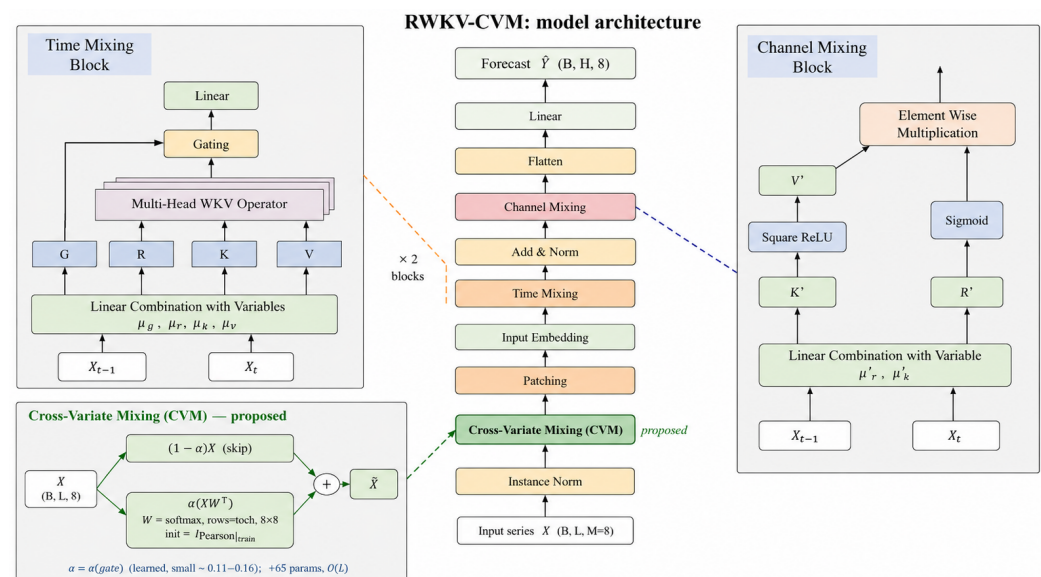


Figure 2. The architecture of the proposed RWKV-CVM. After per-variate instance normalization, the gated CVM module (bottom-left, “proposed”) mixes the $M = 8$ variates via a row-stochastic matrix $\mathbf{W}$ (initialized from the train-set absolute correlations) under a learned gate α ; the mixed series is then patched and each variate is processed independently by the RWKV-TS backbone. The module adds 65 parameters and preserves $O(L)$ complexity. Note that the CVM is applied to the normalized input series *before* patching.

Let $\mathbf{X} \in \mathbb{R}^{L \times M}$ denote the instance-normalized multivariate input. The CVM output is

$$\tilde{\mathbf{X}} = (1 - \alpha) \mathbf{X} + \alpha \mathbf{X} \mathbf{W}^\top \quad (6)$$

where $\mathbf{W} \in \mathbb{R}^{M \times M}$ is a learnable row-stochastic mixing matrix (each row sums to one) and $\alpha \in (0, 1)$ is a learnable scalar gate that controls the strength of Cross-Variate Mixing. The skip term $(1 - \alpha) \mathbf{X}$ ensures that each variate retains its own information, while $\alpha \mathbf{X} \mathbf{W}^\top$ injects a weighted combination of the other variates. The row-stochasticity constraint on $\mathbf{W}$ is enforced through a softmax normalization applied to unconstrained parameters, and the gate α is parameterized via a sigmoid applied to a learnable scalar. The mixed series $\tilde{\mathbf{X}}$ is then patched and processed by the channel-independent RWKV-TS backbone exactly as it is above.

Correlation-Based Initialization

The initialization of the mixing matrix $\mathbf{W}$ influences the convergence behavior and final performance of the CVM module. We adopt a single, data-driven *correlation-based* initialization (CVM-corr). The Pearson correlation coefficient is computed between all pairs

of variates on the training data only, and the absolute correlation matrix is row-normalized to produce the initial $\mathbf{W}_0$:

$$W_{0,ij} = \frac{|\rho_{ij}|}{\sum_{k=1}^M |\rho_{ik}|} \quad (7)$$

where ρ_{ij} is the Pearson correlation between variates i and j . This provides the model with a data-driven prior about which variates are likely to share useful information, while the learned gate α and the subsequent fine-tuning of $\mathbf{W}$ let the model attenuate or refine these couplings during training.

The complete RWKV-CVM architecture is summarized in Algorithm 1 and Figure 2.

3.3. Training Objective

In contrast to the original single-target formulation, which optimized only the Zone 3 loss, we adopt a joint multivariate objective so that the effect of cross-variate sharing on every covariate can be measured. The training loss is the mean squared error averaged over all M forecast channels:

$$\mathcal{L} = \frac{1}{HM} \sum_{t=1}^H \sum_{m=1}^M (y_{t,m} - \hat{y}_{t,m})^2. \quad (8)$$

This joint loss lets the model benefit from—and be evaluated on—cross-variate information across all targets rather than a single zone, directly addressing the question of how the objective affects accuracy for the remaining covariates.

4. Experiment

4.1. Dataset

The experiments are conducted on the power consumption dataset of the distribution network in the city of Tétouan in northern Morocco [31]. The dataset comprises 52,416 observations recorded at 10 min intervals throughout the year 2017, covering the three distribution zones of Quads (Zone 1), Smir (Zone 2), and Boussafou (Zone 3). Each observation includes nine features: three meteorological variables (Temperature, Humidity, WindSpeed), two solar irradiance measures (GeneralDiffuseFlows, DiffuseFlows), and three zonal power consumption values. Following the protocol of Wang et al. [3] for the input features, but in contrast to their single-target setup, all three zones are forecast jointly. To assess generalization beyond a single localized grid, we additionally use two canonical long-term-forecasting benchmarks—ETTh1 (electricity transformer temperature, hourly) and Weather—under a 10% few-shot protocol in which only the first 10% of each training split is used for training [15,16].

Figure 3 displays the full-year power consumption profiles for the three zones. Zone 1 consistently exhibits the highest consumption (typically 30,000–45,000 kW), followed by Zone 3 (15,000–35,000 kW) and Zone 2 (14,000–30,000 kW). All three zones exhibit clear seasonal patterns, with consumption peaks during the summer months (July–August) driven by air-conditioning demand and noticeable daily oscillations reflecting diurnal load patterns.

Figure 4 zooms in to a representative week (12–18 June 2017) at the full 10 min resolution. The daily consumption cycle is evident: consumption drops to its minimum in the early morning hours (around 4–6 AM), rises sharply during the daytime, and peaks in the late evening. Importantly, the three zones exhibit highly similar temporal patterns with consistent lead-lag behavior—Zone 1 consumption often rises slightly before Zone 3 consumption, suggesting that Zone 1 may serve as a leading indicator for Zone 3 forecasting.

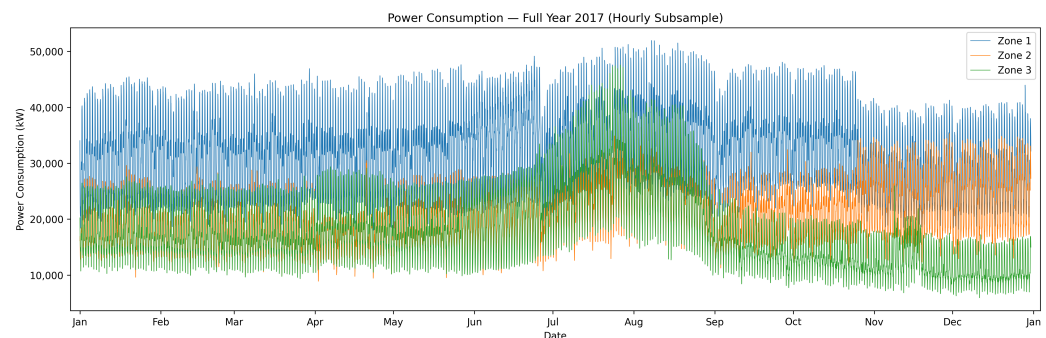


Figure 3. Power consumption profiles for the three distribution zones over the full year 2017 (hourly subsample). Seasonal patterns and inter-zone co-movement are clearly visible.

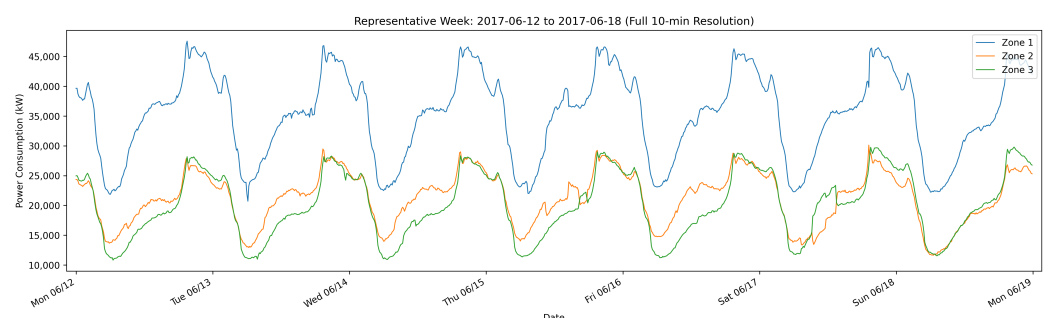


Figure 4. Representative week (12–18 June 2017) at full 10 min resolution, showing the daily consumption cycle and inter-zone co-movement patterns.

4.2. Variable Selection and Correlation Analysis

Figure 5 presents the Pearson correlation matrix for all eight input variables. The strongest correlations with Zone 3 are observed for Zone 1 ($\rho = 0.75$), Zone 2 ($\rho = 0.57$), and Temperature ($\rho = 0.49$). Wind speed also shows a moderate positive correlation ($\rho = 0.28$), while Humidity exhibits a weak negative correlation ($\rho = -0.23$). The two solar irradiance variables (GeneralDiffuseFlows and DiffuseFlows) show negligible correlation with Zone 3 ($\rho = 0.06$ and -0.04 , respectively). These correlations provide an empirical basis for the correlation-based initialization strategy of the CVM mixing matrix (Section 3).

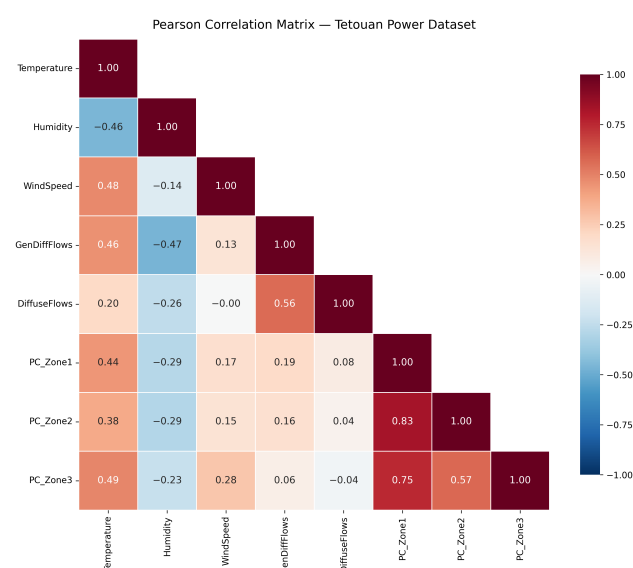


Figure 5. Pearson correlation matrix for the Tetouan power consumption dataset. Zone 1 and Temperature are the variables most strongly correlated with Zone 3.

To complement the linear correlation analysis, Figure 6 shows the Random Forest feature importance scores for predicting Zone 3. Zone 1 dominates with an importance of 0.582, confirming its role as the primary predictor. Zone 2 (0.137) and Temperature (0.135) follow, while the remaining variables each contribute less than 0.05. This non-linear importance ranking is consistent with the Pearson correlations and further supports the hypothesis that cross-variate information—particularly from neighboring zones—is highly relevant for Zone 3 forecasting.

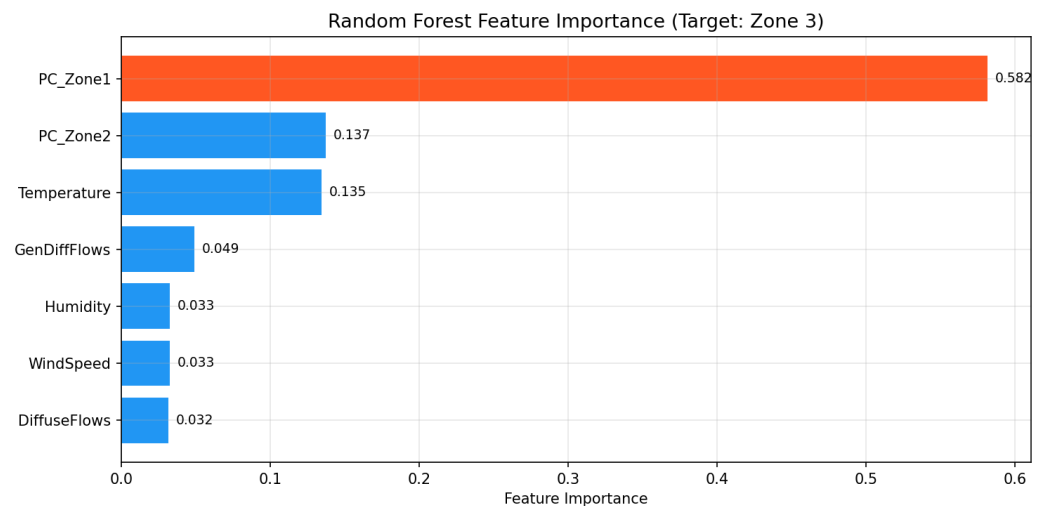


Figure 6. Random Forest feature importance for predicting Zone 3 power consumption. Zone 1 is by far the most important predictor. The bar colors are used only to distinguish the individual features; importance is encoded by the bar length.

4.3. Cross-Correlation and Lead-Lag Analysis

While Pearson correlation measures instantaneous (lag-zero) linear dependence, cross-correlation analysis reveals the temporal lead-lag structure between variates, which is central to the motivation of the CVM module and the recent LIFT approach [30].

Figure 7 displays the cross-correlation function (CCF) between Zone 1 and Zone 3. The maximum cross-correlation $|R(\tau^*)| = 0.7554$ occurs at an optimal lag of $\tau^* = 2$ steps (20 min), indicating that Zone 1 consumption leads Zone 3 by approximately 20 min. This short lead time reflects the spatial proximity of the two distribution zones.

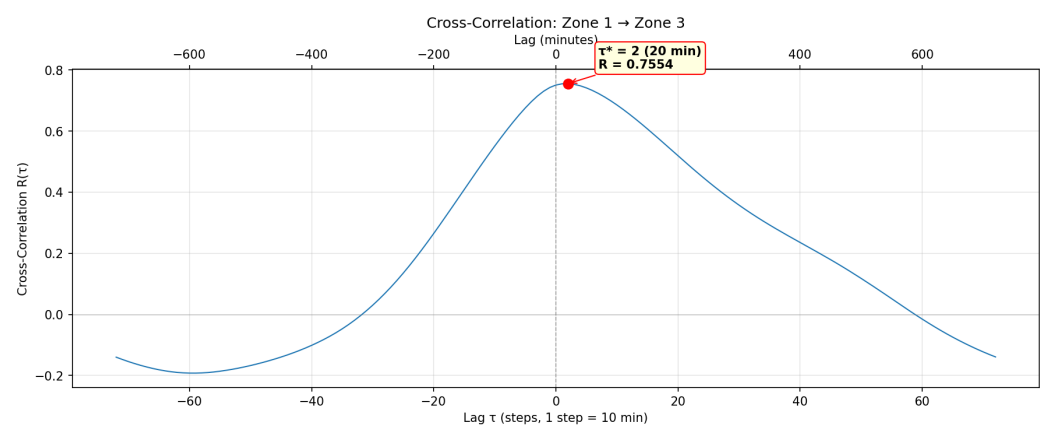


Figure 7. Cross-correlation function from Zone 1 to Zone 3. The peak at $\tau^* = 2$ (20 min) indicates that Zone 1 leads Zone 3 by two time steps.

Figure 8 extends the analysis to all seven input variates with respect to Zone 3. Zone 1 and Zone 2 exhibit the strongest cross-correlations and the shortest optimal lags ($\tau^* = 2$ steps each). Temperature leads Zone 3 by 28 steps (~ 4.7 h) with $|R| = 0.583$, consistent with the thermal inertia effect, where rising temperatures drive subsequent increases in cooling-related electricity demand. The solar irradiance variables (GeneralDiffuseFlows and DiffuseFlows) show longer lags (35–45 steps) and moderate correlations, reflecting the indirect influence of solar radiation on consumption through temperature.

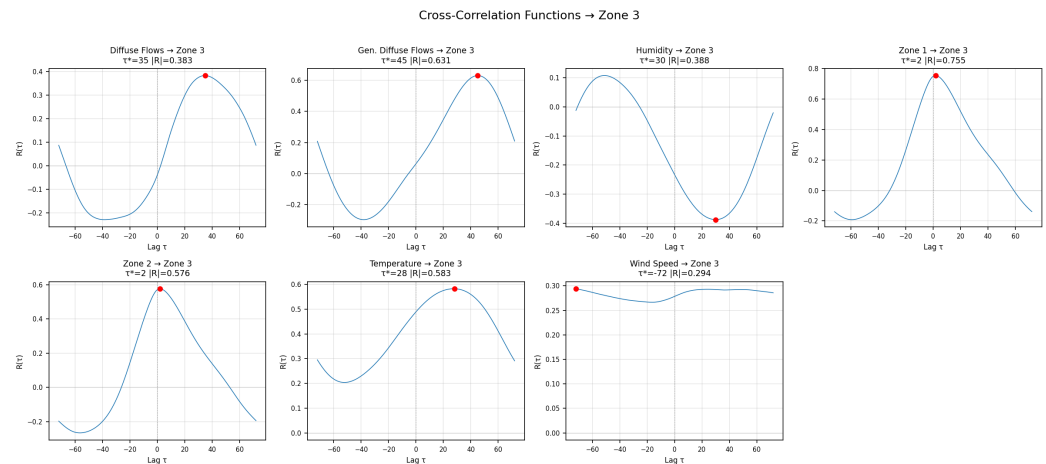


Figure 8. Cross-correlation functions from all input variates to Zone 3. Optimal lags τ^* and peak absolute correlations $|R|$ are annotated for each pair.

A comprehensive view of the pairwise cross-correlation structure is provided in Figures 9 and 10. Figure 9 shows the maximum absolute cross-correlation $|R(\tau^*)|$ for all variable pairs. The three consumption zones form a strongly correlated cluster ($|R| > 0.57$), while weather variables show weaker but non-negligible cross-dependencies. Figure 10 maps the optimal lag τ^* for each pair, revealing that the Zone 3 row has mostly negative values (Zone 3 lags behind other variables), which further validates the potential of using leading indicators for Zone 3 forecasting.

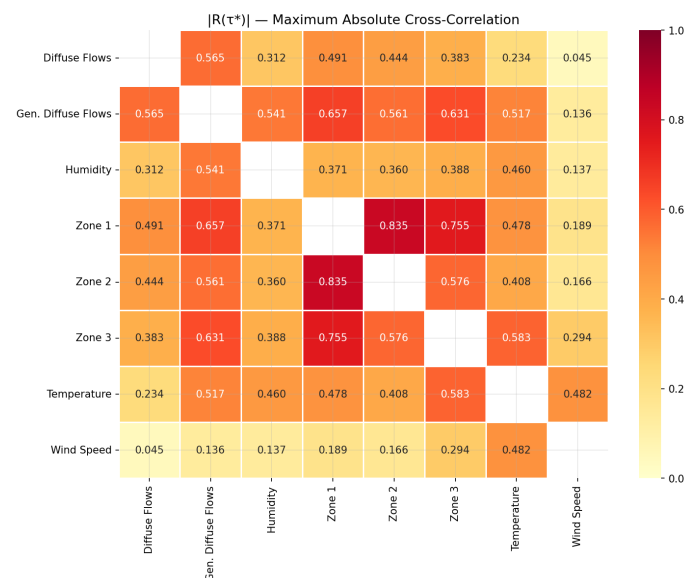


Figure 9. Heatmap of maximum absolute cross-correlation $|R(\tau^*)|$ between all variable pairs.

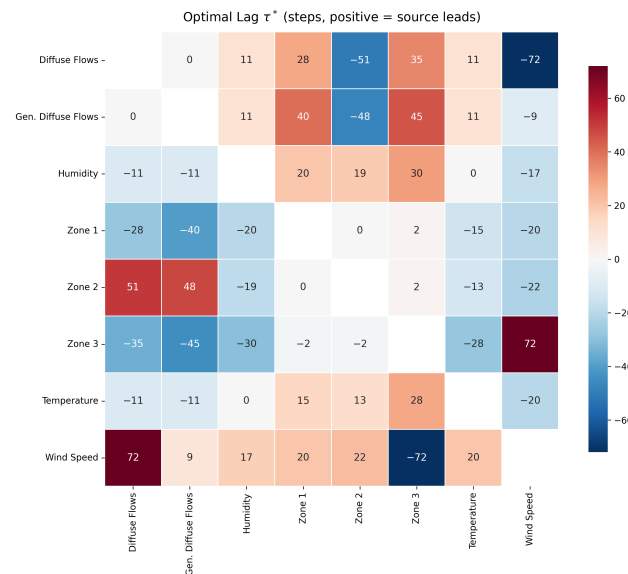


Figure 10. Heatmap of optimal lag τ^* (in 10 min steps) for all variable pairs. Positive values indicate that the source (row) leads the target (column).

4.4. Stationarity of Lead-Lag Relationships

An important consideration for any channel-dependent model is whether the inter-variate dependencies remain stable over time. Figure 11 tracks the optimal lag τ^* over the full year using sliding 24 h windows for three representative pairs. The Zone 1 $\rightarrow$ Zone 3 relationship is remarkably stable, with a mean lag of 1.9 ± 1.4 steps and minimal seasonal variation. In contrast, the GeneralDiffuseFlows $\rightarrow$ Zone 3 relationship exhibits higher variability (mean 35.4 ± 17.5 steps), particularly during the summer months when solar patterns become more erratic. The Temperature $\rightarrow$ Zone 3 lag shows intermediate stability (mean 22.0 ± 19.7 steps).



Figure 11. Temporal evolution of the optimal lead-lag τ^* over 2017 using 24 h sliding windows, for Zone 1 $\rightarrow$ Zone 3 (top), GeneralDiffuseFlows $\rightarrow$ Zone 3 (middle), and Temperature $\rightarrow$ Zone 3 (bottom). The red dashed line indicates the mean lag; the shaded area represents ± 1 standard deviation.

This analysis motivates two key design decisions in the CVM module: (i) the use of a *static* mixing matrix $\mathbf{W}$ —justified by the overall stationarity of the most predictive relationships (Zone 1, Zone 2)—combined with (ii) a learnable gate α that can attenuate the influence of less stable variates. The choice of a static $\mathbf{W}$ is a deliberate trade-off: the most predictive lead-lag relationship (Zone 1→Zone 3) is stationary, whereas a fully time-variant scheme would add parameters and could re-introduce the spurious, non-stationary couplings that channel-dependent models are prone to overfit. As shown later (Section 5.5), the learned gate α and the matrix $\mathbf{W}$ do shift with the forecast horizon, providing a degree of adaptivity; a fully dynamic, time-variant mixing scheme is left for future work.

Augmented Dickey–Fuller Stationarity Test

To complement the sliding-window lead-lag analysis with a formal test, we apply the Augmented Dickey–Fuller (ADF) test to each channel. As reported in Table 1, all eight channels reject the unit-root null hypothesis at the 1% level, indicating that the series are stationary and that a static correlation prior is a reasonable modeling choice.

Table 1. Augmented Dickey–Fuller (ADF) stationarity test for all channels of the Tetouan dataset. All channels reject the unit-root null hypothesis (stationary).

Channel	ADF Statistic	<i>p</i> -Value	Stationary
Temperature	−8.18	8.1×10^{-13}	Yes
Humidity	−17.73	3.4×10^{-30}	Yes
WindSpeed	−5.68	8.7×10^{-7}	Yes
GeneralDiffuseFlows	−29.31	$<10^{-20}$	Yes
DiffuseFlows	−29.33	$<10^{-20}$	Yes
PowerConsumption Zone 1	−28.86	$<10^{-20}$	Yes
PowerConsumption Zone 2	−22.21	$<10^{-20}$	Yes
PowerConsumption Zone 3	−20.51	$<10^{-20}$	Yes

4.5. Data Normalization and Leakage Check

Following Wang et al. [3], the data are split chronologically into training (70%), validation (10%), and test (20%) sets to prevent data leakage. To remove ambiguity at the boundaries, every input–horizon window is fully contained within a single split; no window crosses a split boundary, and the exact split indices are released (Data and Code Availability). We employ instance normalization [32] rather than global Z-Score standardization. For each variate, the mean and standard deviation are computed from the input window and used to normalize the series to zero mean and unit variance. These statistics are stored and reapplied through a reverse normalization step on the output predictions, which helps mitigate the distribution shift between training and test data.

Because the CVM correlation prior (Equation (7)) is estimated from data, we verify that it does not leak test-set information. The prior is computed on the training partition only. Quantifying the drift between the training-set correlation matrix and the evaluation-set correlation matrix gives a mean absolute entry difference of 0.118 (maximum 0.386, Frobenius norm 1.31), confirming that the prior reflects training-set structure rather than test-set structure and that the gate/fine-tuning subsequently adapts it.

4.6. Methods for Comparison

A central concern with the original submission is that the baselines were taken directly from Wang et al. [3] under a different normalization scheme, which prevents a fair comparison. To address this, we re-execute three strong, modern baselines under the *same* pipeline, normalization, seeds and horizons as RWKV-CVM, and we use these re-run results—rather than directly cited numbers—for all quantitative claims:

DLinear [20]: a simple yet strong linear decomposition baseline that has been shown to match or exceed many Transformer models.

iTransformer [28]: an inverted Transformer that attends across variate tokens, representing the channel-dependent paradigm.

RWKV-TS (A0 Baseline): The channel-independent RWKV-TS model [24] without CVM, serving as our ablation baseline. RWKV-CVM(corr) differs from this backbone only by the 65-parameter CVM module, so the RWKV-TS vs. RWKV-CVM comparison isolates the contribution of Cross-Variate Mixing.

For reference, the prior best result on this dataset [3] used a two-layer LSTM-Informer ensemble. Because those numbers were produced under a different normalization and a single-target (Zone 3) setup, they are not directly comparable to ours, and we therefore do not tabulate them as a head-to-head comparison; the re-run baselines above provide the controlled comparison instead.

4.7. Experimental Configuration

We compare RWKV-CVM(corr) against the re-run DLinear, iTransformer and RWKV-TS baselines. RWKV-TS is the channel-independent backbone (the A0 baseline of the module ablation), and RWKV-CVM(corr) adds only the gated CVM module.

All RWKV configurations share the same backbone hyperparameters: two encoder layers, two attention heads, hidden dimension $D = 128$, patch size $P = 16$, stride $S = 8$, and dropout rate of 0.1. These follow the published RWKV-TS defaults [24]; the same values are used for the backbone and the CVM variant so that any difference is attributable to the module rather than to tuning. The input sequence length is $L = 288$ steps (48 h) for Tetouan, and five prediction horizons are evaluated: $H \in \{24, 48, 144, 288, 432\}$ steps (4, 8, 24, 48 and 72 h). For the ETTh1 and Weather few-shot benchmarks, the standard long-term-forecasting horizons $\{96, 192, 336, 720\}$ are used.

Training is performed with the AdamW optimizer [33] using an initial learning rate of 10^{-4} , weight decay of 0.05, cosine learning rate scheduling, and a batch size of 64. Early stopping with a patience of seven epochs is applied based on validation loss. All experiments are repeated over three random seeds (42, 43, 44). The models are implemented in Python with PyTorch and trained on a single NVIDIA Tesla T4 GPU (NVIDIA Corporation, Santa Clara, CA, USA) provided by the Kaggle platform; the exact versions of all software libraries are listed in the released code repository (see the Data Availability Statement). We report the three-seed mean for every metric, and 95% confidence intervals are shown as error bars in the result figures and tabulated in full in the supplementary spreadsheet.

4.8. Evaluation Metrics

This paper selects MSE and MAE to contrast the precision of different algorithms, following the same metrics used by Wang et al. [3]. MSE is a measure of the average squared difference between the predicted and true values; MAE measures the average absolute deviation. The formulas are

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (9)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (10)$$

where n represents the number of prediction time points, $\hat{y}_i$ represents the result of the i -th model prediction, and y_i represents the true value. For statistical comparison, we additionally apply a Diebold–Mariano test with Holm correction at the window level; we

treat these as diagnostics and footnote the cells where the pooled-window test points the opposite way to the three-seed mean.

5. Results and Analysis

All values reported below are three-seed (42/43/44) means under the unified harness of Section 4.7; the lowest value per column is shown in **bold**. Only the correlation-initialized variant of CVM is considered. We deliberately avoid blanket-superiority language: RWKV-CVM is best on the horizon *average* of each dataset, by small margins, and a baseline is preferable on several individual horizons, which we state explicitly.

5.1. Joint All-Zone Comparison on Tetouan

Table 2 (MSE) and Table 3 (MAE) report the three-seed results on the Tetouan dataset with all three zones forecast jointly, and Figures 12 and 13 visualize them with 95% confidence-interval error bars. Averaged over the five horizons, RWKV-CVM attains the lowest mean MSE (0.0427) and the lowest mean MAE (0.1340), narrowly ahead of iTransformer (MSE 0.0429), its RWKV-TS backbone (MSE 0.0435) and DLinear (MSE 0.0495). The margins are small and the ranking is not uniform across horizons: iTransformer attains the best MSE at $H = 48$ (0.0257) and $H = 432$ (0.0681) and the best MAE at $H = 432$ (0.1764), while RWKV-TS attains the best MAE at the two shortest horizons. RWKV-CVM is strongest at the operationally relevant 24 h ($H = 144$) and 48 h ($H = 288$) horizons, where it gives the best MSE and MAE.

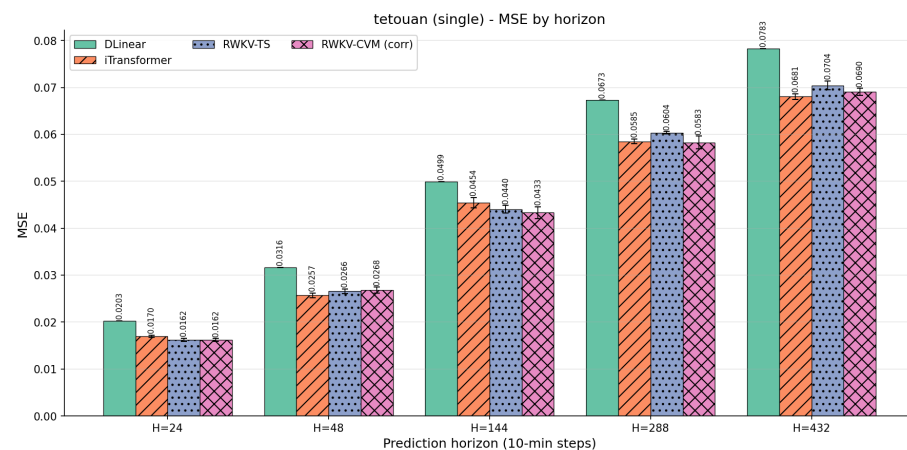


Figure 12. Tetouan all-zone (joint) MSE versus horizon, averaged over the three zones (three-seed mean; error bars are 95% confidence intervals). RWKV-CVM is best on the horizon average; iTransformer wins at $H = 48$ and $H = 432$.

Table 2. Tetouan all-zone (joint) three-seed mean MSE. Lower is better; best per column in **bold**.

Model	H = 24 (4 h)	H = 48 (8 h)	H = 144 (24 h)	H = 288 (48 h)	H = 432 (72 h)	Avg
DLinear	0.0203	0.0316	0.0499	0.0673	0.0783	0.0495
iTransformer	0.0170	0.0257	0.0454	0.0585	0.0681	0.0429
RWKV-TS	0.0162	0.0266	0.0440	0.0604	0.0704	0.0435
RWKV-CVM (corr)	0.0162	0.0268	0.0433	0.0583	0.0690	0.0427

Table 3. Tetouan all-zones (joint) three-seed mean MAE. Lower is better; best per column in **bold**.

Model	H = 24	H = 48	H = 144	H = 288	H = 432	Avg
DLinear	0.0925	0.1156	0.1438	0.1699	0.1846	0.1413
iTransformer	0.0881	0.1085	0.1426	0.1622	0.1764	0.1356
RWKV-TS	0.0851	0.1081	0.1378	0.1625	0.1785	0.1344
RWKV-CVM (corr)	0.0857	0.1097	0.1371	0.1601	0.1776	0.1340

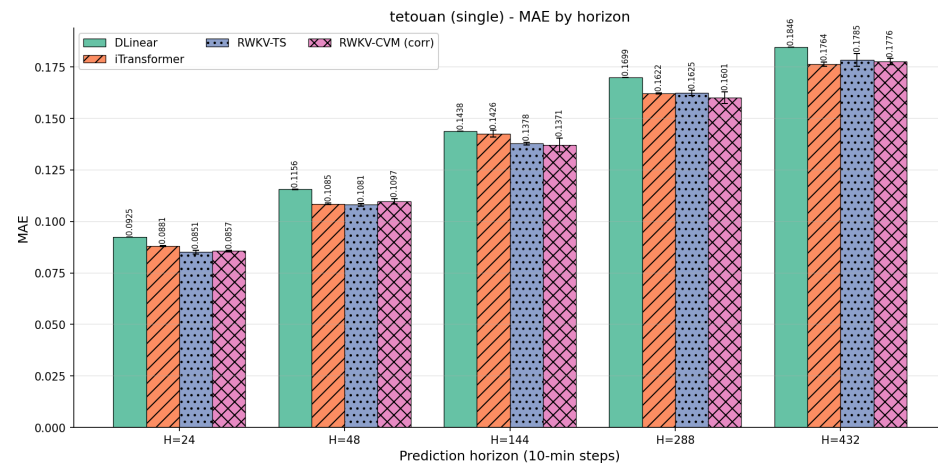


Figure 13. Tetouan all-zones (joint) MAE versus horizon, averaged over the three zones (three-seed mean; 95% confidence-interval error bars).

5.2. Module Ablation: Effect of the CVM

Because RWKV-CVM(corr) differs from the RWKV-TS backbone only by the 65-parameter CVM module, the RWKV-TS (A0) vs. RWKV-CVM(corr) rows of Table 2 constitute a controlled, multi-seed ablation of the module; Table 4 isolates them. Adding the CVM lowers the horizon-average MSE from 0.0435 to 0.0427 (a 1.8% relative reduction). The effect is concentrated at the longer horizons—the largest gains are at $H = 288$ (−3.5%) and $H = 432$ (−2.0%)—is essentially a tie at $H = 24$, and is marginally negative at $H = 48$. This pattern is consistent with the learned gate increasing with horizon (Section 5.5): the cross-variate branch is a small correction that the model relies on more for longer forecasts.

Table 4. Module ablation on Tetouan all-zones (three-seed mean MSE): the channel-independent backbone (RWKV-TS, A0) versus RWKV-CVM(corr), which adds only the 65-parameter CVM module. “ Δ ” is the relative change in CVM with respect to the backbone (negative = improvement). The lower (better) value in each column is shown in bold.

Configuration	H = 24	H = 48	H = 144	H = 288	H = 432	Avg
RWKV-TS (A0)	0.0162	0.0266	0.0440	0.0604	0.0704	0.0435
RWKV-CVM (corr)	0.0162	0.0268	0.0433	0.0583	0.0690	0.0427
Δ (%)	≈ 0	+0.8	−1.6	−3.5	−2.0	−1.8

5.3. Generalization: ETTh1 and Weather Few-Shot

To test whether the module generalizes beyond a single localized grid, we evaluate the same four models on the ETTh1 and Weather benchmarks under a 10% few-shot protocol. Tables 5 and 6 (MSE) report the three-seed means, and Figures 14 and 15 visualize them.

On ETTh1 (Table 5), the RWKV family clearly outperforms DLinear and iTransformer, and RWKV-CVM attains the best horizon-average MSE (0.6396), marginally ahead of its backbone (0.6411). The per-horizon picture is mixed: RWKV-TS is best at $H = 96$, DLinear is best at $H = 192$ and $H = 336$, and RWKV-CVM is best at $H = 720$. On Weather (Table 6), RWKV-CVM attains the best average MSE (0.2499) and the best MSE at $H = 96$, $H = 192$ and $H = 336$, while DLinear is best at $H = 720$. In MAE terms, RWKV-CVM and RWKV-TS are essentially tied on Weather (average MAE 0.2838 vs. 0.2837, the backbone is marginally better), underscoring that the advantage is small.

Table 5. ETTh1 10% few-shot, three-seed mean MSE (top) and MAE (bottom). Best per column in **bold**.

Model	H = 96	H = 192	H = 336	H = 720	Avg
<i>MSE</i>					
DLinear	0.5483	0.5761	0.6674	0.9168	0.6771
iTransformer	0.6378	0.7370	0.7708	0.9201	0.7664
RWKV-TS	0.4797	0.6021	0.6867	0.7959	0.6411
RWKV-CVM (corr)	0.4832	0.5926	0.6944	0.7883	0.6396
<i>MAE</i>					
DLinear	0.5055	0.5202	0.5627	0.6713	0.5650
iTransformer	0.5389	0.5773	0.5978	0.6731	0.5968
RWKV-TS	0.4654	0.5347	0.5799	0.6275	0.5519
RWKV-CVM (corr)	0.4665	0.5274	0.5802	0.6247	0.5497

Table 6. Weather 10% few-shot, three-seed mean MSE (top) and MAE (bottom). Best per column in **bold**.

Model	H = 96	H = 192	H = 336	H = 720	Avg
<i>MSE</i>					
DLinear	0.1806	0.2234	0.2694	0.3392	0.2532
iTransformer	0.1916	0.2426	0.2878	0.3622	0.2711
RWKV-TS	0.1839	0.2186	0.2665	0.3408	0.2525
RWKV-CVM (corr)	0.1760	0.2177	0.2651	0.3409	0.2499
<i>MAE</i>					
DLinear	0.2320	0.2665	0.3007	0.3485	0.2869
iTransformer	0.2427	0.2862	0.3208	0.3712	0.3052
RWKV-TS	0.2241	0.2615	0.2983	0.3510	0.2837
RWKV-CVM (corr)	0.2248	0.2606	0.2975	0.3522	0.2838

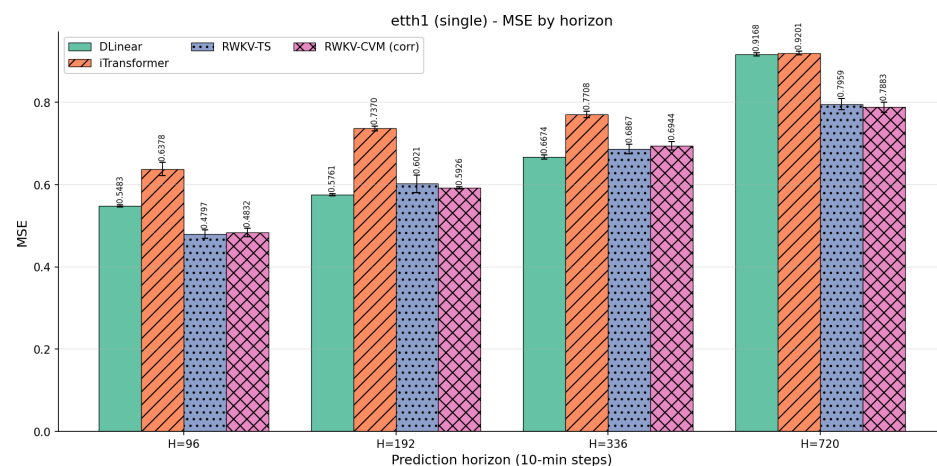
**Figure 14.** ETTh1 10% few-shot MSE versus horizon (three-seed mean; 95% confidence-interval error bars). The RWKV family outperforms DLinear and iTransformer.

Table 7 summarizes the horizon-average MSE across the three datasets. RWKV-CVM attains the lowest average MSE on all three, but the margins over its backbone and over the strongest baseline are small, and the advantage does not extend to every horizon. Diebold–Mariano diagnostics point the opposite way to the three-seed mean at ETTh1 $H = 192/336$, Weather $H = 192/336$ and Tetouan all-zones $H = 48/144$; with only three seeds, the per-seed Wilcoxon test is uninformative, so we rely on the three-seed mean with confidence intervals together with the Diebold–Mariano direction and report disagreements transparently.

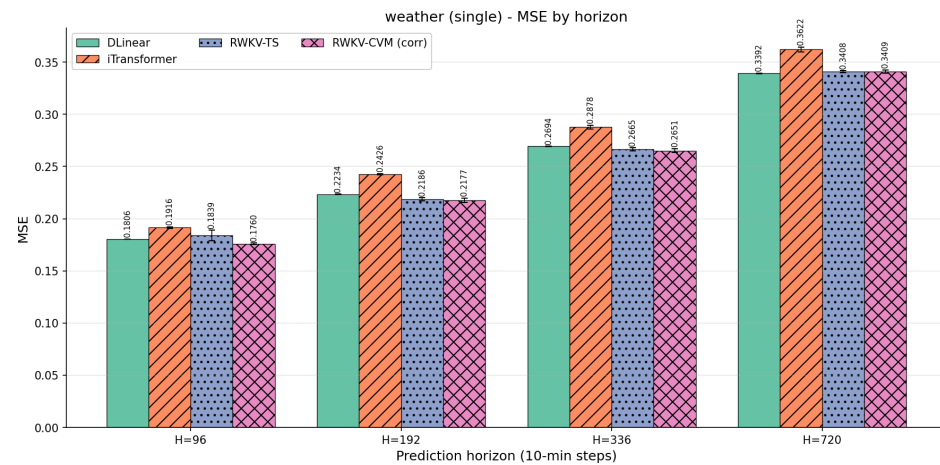


Figure 15. Weather 10% few-shot MSE versus horizon (three-seed mean; 95% confidence-interval error bars). RWKV-CVM is best at $H = 96/192/336$.

Table 7. Cross-dataset horizon-average MSE (three-seed mean; lower is better). Best per column in **bold**.

Model	Tetouan All-Zones	ETTh1 Few-Shot	Weather Few-Shot
DLinear	0.0495	0.6771	0.2532
iTransformer	0.0429	0.7664	0.2711
RWKV-TS	0.0435	0.6411	0.2525
RWKV-CVM (corr)	0.0427	0.6396	0.2499

5.4. Per-Zone Transferability

Forecasting all three zones jointly lets us assess whether cross-variant sharing transfers across target variables. Table 8 reports the per-zone horizon-average MSE, and Figure 16 shows the full per-zone curves. The picture is *selective* rather than uniform. RWKV-CVM improves over its RWKV-TS backbone on all three zones on average, but it only wins outright on Zone 1 (where it essentially ties the backbone), whereas iTransformer is best on Zone 2 and DLinear is best on Zone 3. The benefit of Cross-Variate Mixing is therefore concentrated on the zones that have strong, stable couplings (Zone 1 and Zone 2, which the learned matrix couples most strongly; Section 5.5), and it does not improve the more autonomous Zone 3—consistent with Zone 3 being a lagging rather than a leading indicator (Section 4.4).

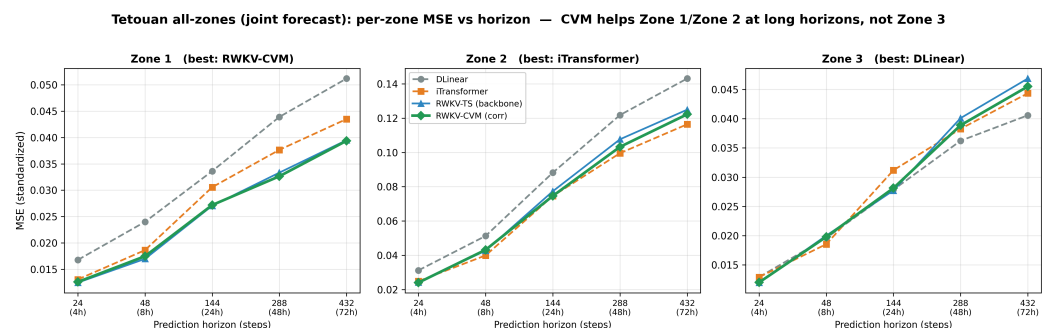


Figure 16. Per-zone MSE versus horizon on Tetouan all-zones (three-seed mean). Cross-Variate Mixing helps Zone 1 and Zone 2 (most at the longer horizons) but not the more autonomous Zone 3, where DLinear is best.

Table 8. Per-zone horizon-average MSE on Tetouan all-zones (three-seed mean over five horizons). Best per zone in **bold**.

Target Zone	DLinear	iTransformer	RWKV-TS	RWKV-CVM (corr)
Zone 1	0.03390	0.02868	0.02586	0.02585
Zone 2	0.08711	0.07106	0.07537	0.07347
Zone 3	0.02743	0.02905	0.02933	0.02886

5.5. What the Module Learns: Gate and Mixing Weights

To interpret the module, we examine the learned gate α and mixing matrix $\mathbf{W}$. Figure 17 shows that the gate rises with the forecast horizon: across horizons, the converged value increases from $\alpha \approx 0.115$ at $H = 24$ to $\alpha \approx 0.160$ at $H = 432$, and within a run, it grows over epochs at the long horizons (to ~ 0.19 at $H = 288/432$) while decaying slightly at $H = 24$. On the ETTh1 and Weather benchmarks, the gate is smaller and roughly flat ($\alpha \approx 0.12$). The interpretation is that the cross-variate branch is a small ($\sim 12\%$) residual correction that the model relies on more for longer forecasts—consistent with the gains being concentrated at the longer horizons (Section 5.2).

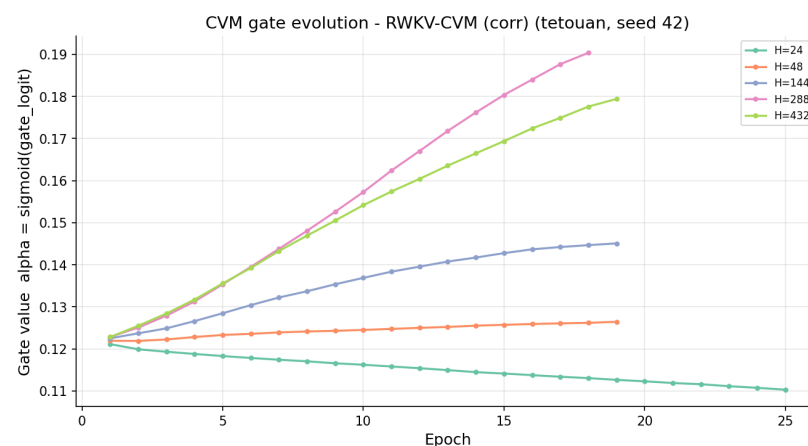


Figure 17. Evolution of the gate $\alpha = \sigma(\text{gate})$ during training on Tetouan (seed 42). α increases with the forecast horizon and grows over epochs at the longer horizons, indicating that the model leans more on Cross-Variate Mixing for longer forecasts.

Figure 18 shows the learned matrix $\mathbf{W}$ across horizons. It is mildly diagonal-dominant (self-weight ≈ 0.27 – 0.35 , decreasing as the horizon grows), and the dominant off-diagonal couplings are between Zone 1 and Zone 2 (≈ 0.37 – 0.39), which explains why the per-zone gains concentrate on those two zones. Table 9 reports the Zone 3 output row of $\mathbf{W}$ across all five horizons (answering the request for the full-horizon weights): the self-weight on Zone 3 decreases from 0.35 at $H = 24$ to 0.27 at $H = 432$, while the weights on Zone 1, Zone 2 and Temperature remain the largest cross-variate contributors—in line with the correlation and importance analyses of Section 4.

Table 9. Learned CVM mixing weights for the Zone 3 output row across all five horizons (read from Figure 18; rows sum to 1). Zone 1, Zone 2 and Temperature remain the dominant cross-variate contributors; the Zone 3 self-weight decreases with horizon.

Horizon	Temp	Hum	Wind	GDF	DF	Z1	Z2	Z3
H = 24	0.09	0.07	0.06	0.00	0.03	0.23	0.17	0.35
H = 48	0.09	0.07	0.07	0.00	0.03	0.23	0.19	0.33
H = 144	0.09	0.08	0.06	0.00	0.03	0.20	0.19	0.32
H = 288	0.12	0.09	0.11	0.00	0.04	0.19	0.18	0.27
H = 432	0.13	0.09	0.11	0.00	0.03	0.20	0.17	0.27

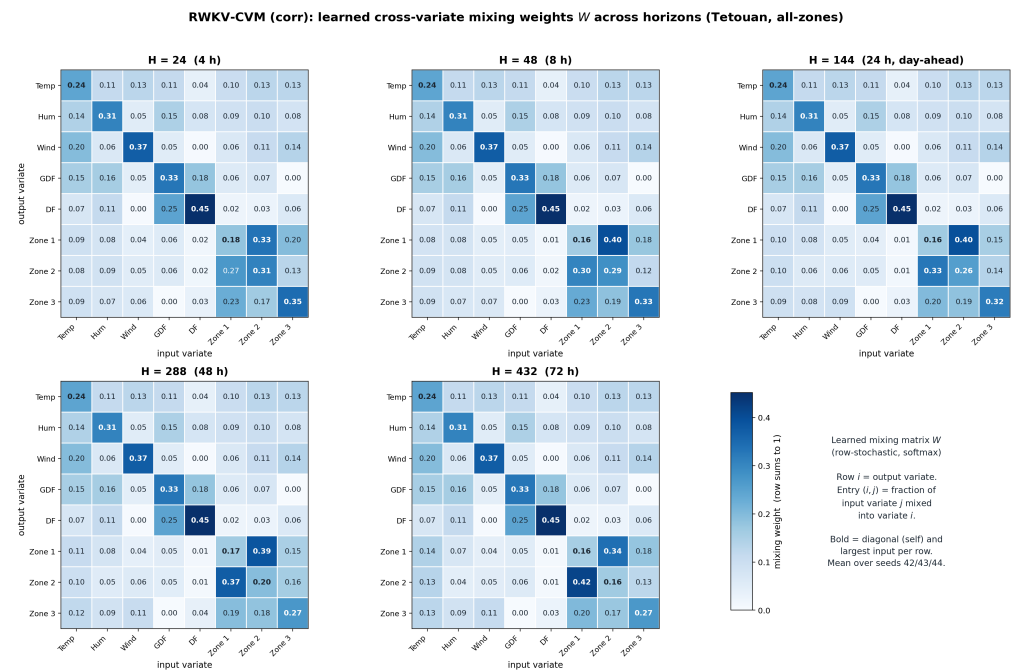


Figure 18. Learned row-stochastic mixing matrix W (mean over seeds) at each horizon on Tetouan all-zones. Row i is an output variate; entry (i, j) is the fraction of input variate j mixed into variate i . The matrix is mildly diagonal-dominant with a strong Zone 1 $\leftrightarrow$ Zone 2 coupling.

5.6. Computational Overhead

An important practical consideration is the cost of the CVM module. Table 10 compares parameter count, inference latency and FLOPs of the four models (Tetouan, $L = 288$, measured at a representative horizon). The CVM module adds only 65 parameters ($M^2 + 1 = 8 \times 8 + 1$) over RWKV-TS and has identical FLOPs, so the module itself is essentially free. At the same time, the RWKV family is 3–14 $\times$ larger and slower than DLinear, which matches the RWKV models at the longest horizons (Section 5.3). The contribution of CVM is therefore accuracy on average, not efficiency.

Table 10. Complexity comparison on Tetouan (parameter count, mean inference latency per batch, and FLOPs). The CVM module adds +65 parameters and identical FLOPs over the RWKV-TS backbone.

Model	Params	Latency (ms)	FLOPs
DLinear	73,920	0.61	1.18 M
iTransformer	326,896	1.28	3.08 M
RWKV-TS	1,002,224	4.63	318.7 M
RWKV-CVM (corr)	1,002,289	4.75	318.7 M

Linear Length Complexity

The CVM operation $\tilde{X} = (1 - \alpha)X + \alpha XW^T$ multiplies an $L \times M$ matrix by an $M \times M$ matrix, costing $O(LM^2)$, which is linear in the sequence length L for a fixed number of variates M . The RWKV-TS backbone is $O(L)$ by construction [24]. Their composition is therefore $O(L)$ in the sequence length, and the empirical latency in Table 10 confirms that adding the module leaves runtime essentially unchanged.

5.7. Prediction Quality Visualization

Figure 19 shows the day-ahead ($H = 144$, 24 h) forecasts of RWKV-CVM for all three zones on a median-error window, and Figure 20 shows the two-day-ahead ($H = 288$, 48 h) forecasts. The model tracks the diurnal cycle of each zone well at both horizons, with the

largest deviations occurring around sharp evening peaks and troughs. Per-zone forecasts at the 10th, 50th and 90th error percentiles—including the high-error (90th-percentile) windows that probe extreme behavior—are provided for completeness in Appendix A.

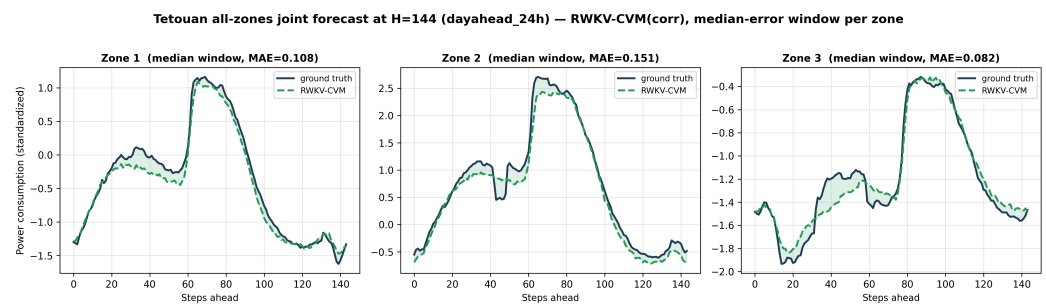


Figure 19. Day-ahead ($H = 144$, 24h) forecasts of RWKV-CVM for the three zones on a median-error window (standardized units). The model captures the daily cycle of each zone.

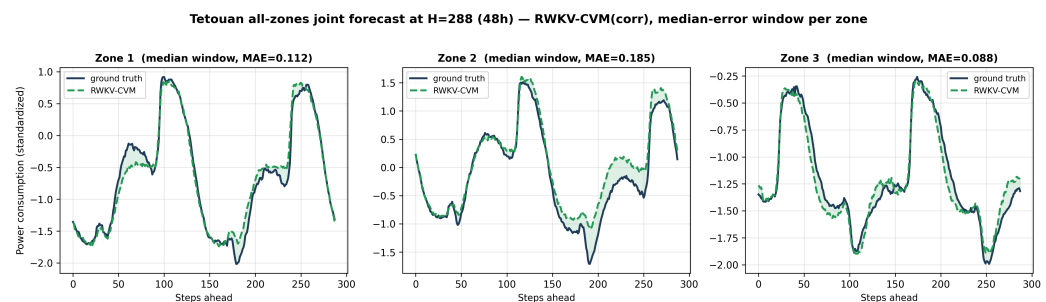


Figure 20. Two-day-ahead ($H = 288$, 48h) forecasts of RWKV-CVM for the three zones on a median-error window (standardized units).

5.8. Why Gated Cross-Variate Mixing Helps

The empirical results admit a simple interpretation that also clarifies why the gating and the row-stochastic constraint are appropriate.

Interpolating between channel independence and dependence. Equation (6) interpolates between two regimes: at $\alpha = 0$ the model is purely channel-independent (each variate is processed in isolation, including the regime that is robust but discards cross-variate signal), and at $\alpha = 1$, the input of every variate is replaced by a fixed convex combination of all variates (a strong, static channel-dependent regime that risks importing noise from weakly related series). A learned $\alpha \in (0, 1)$ lets the model choose how far to move along this axis. Empirically, the model selects a small $\alpha \approx 0.11$ – 0.19 (Section 5.5); i.e., it keeps most of each variate’s own information and adds only a minor cross-variate correction—which is exactly the behavior that avoids the overfitting that pure channel-dependent methods are prone to, while still recovering some of the signal that pure channel-independent methods discard.

Gradient flow and the role of the gate. The skip path $(1 - \alpha)\mathbf{X}$ guarantees an identity-like gradient route to each variate, so the backbone receives a clean training signal even when the mixing branch is uninformative; the mixing branch only contributes a fraction α of the gradient. The gate therefore acts as a learnable regularizer on cross-variate information: it can shrink towards 0 when mixing hurts (as it nearly does at the shortest horizon) and grow when mixing helps (as at the longer horizons). This is consistent with an information-bottleneck reading in which the model admits cross-variate information only to the extent that it reduces the forecasting loss.

Row-stochasticity as a mass-preserving averaging operator. Constraining each row of $\mathbf{W}$ to sum to one (via softmax) makes the mixing branch an *averaging* operator: each mixed variate is a convex combination of the inputs, so the operator maps the probability simplex to itself and cannot amplify the overall magnitude of the signal—it can only redistribute

it across variates. In particular, if all variates carried the same value, mixing would leave them unchanged. This is a mean-preserving (mass-preserving) property of the operator; we note that it is a property of the normalized representation and not a claim about the physical energy-conservation law of the grid. The constraint keeps the mixing well-conditioned and prevents the module from inflating any single variate, which is helpful given that the variates are on a common normalized scale.

5.9. Limitations and Scope

We scope the claims carefully. First, the advantage of RWKV-CVM is on the horizon *average* of the few-shot and joint all-zones settings; in the *full-data* regime, it is *not* the winner. Under full-data training, DLinear gives the lowest average MSE on the single-target Tetouan-Zone 3 setting (0.0283, with RWKV-CVM at 0.0294, third) and on ETTh1 (0.1011, RWKV-CVM 0.1042, second), while iTransformer is best on the ECL benchmark (0.1498, RWKV-CVM 0.1576, second); RWKV-CVM beats both DLinear and iTransformer on *zero* full-data datasets. We therefore make no full-data superiority claim.

Second, the gains are small, are concentrated at longer horizons, and are selective across zones (Section 5.4): the more autonomous Zone 3 is not improved. Third, several baselines remain to be run: PatchTST and the closest conceptual neighbor, LIFT, are not evaluated here, and a comparison against them is left for future work. Fourth, the study uses a single localized grid plus two general LTSF benchmarks. A macroscopic grid such as ERCOT, an overlay of physical substation topology on the lead-lag and mixing-weight analysis, a placement ablation (mixing the raw input before the encoder versus mixing between layers), a fully time-variant mixing scheme, and a gate-sensitivity sweep over patch length and stride are all natural extensions that we did not perform. We report these openly so that the contribution—a nearly free, interpretable module that improves multivariate load forecasting on average—is not overstated.

6. Conclusions

This paper introduced RWKV-CVM, a lightweight extension of the RWKV-TS architecture for multivariate time series forecasting that incorporates a trainable Cross-Variate Mixing module. The module applies a gated, row-stochastic mixing of the normalized input variates—initialized from the training-set correlations—before the channel-independent backbone, adding only 65 parameters and preserving $O(L)$ complexity.

Under a single unified harness—three random seeds, consistent normalization across all models, and re-executed DLinear, iTransformer and RWKV-TS baselines—the module lowers the horizon-average MSE on all three evaluated settings: the Tetouan dataset forecast jointly for all three zones (0.0427 vs. 0.0435 for the backbone) and the ETTh1 (0.640) and Weather (0.250) few-shot benchmarks. The improvement is modest, is concentrated at the longer (day-ahead and beyond) horizons where the learned gate is largest, and is selective across target zones: it helps the strongly and coupled Zone 1 and Zone 2 but not the more autonomous Zone 3. We make no blanket-superiority claim—on several individual horizons, a baseline is preferable, and in the full-data regime, RWKV-CVM is not the best model—and we report these cases explicitly in Section 5.9. The learned gate and mixing matrix are interpretable and recover the inter-zone and temperature dependencies identified in the exploratory analysis.

Several directions remain open for future work. A fully dynamic, time-variant mixing matrix (rather than the static, gated matrix used here) could better track non-stationary couplings at the longest horizons. Evaluating the CVM module against, and on top of, additional baselines—most notably PatchTST and the closely related lead-lag method LIFT [30]—would situate it more precisely among cross-variate methods. Applying the

module to a macroscopic grid such as ERCOT, overlaying the physical substation topology on the lead-lag and mixing-weight analysis, and running a placement and gate-sensitivity ablation are further natural extensions. Additional per-zone prediction visualizations, including high-error windows, are provided in Appendix A.

Author Contributions: Conceptualization, A.R. and A.E.; methodology, A.R.; software, A.R.; validation, A.R., A.E. and H.Y.; formal analysis, A.R.; investigation, A.R.; writing—original draft preparation, A.R.; writing—review and editing, A.E. and H.Y.; visualization, A.R.; supervision, A.E. and H.Y.; project administration, A.E. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The Tetouan city power consumption dataset used in this study is publicly available on the UCI Machine Learning Repository at <https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city> (accessed on 1 March 2026). The ETTh1 and Weather benchmarks are publicly available from the Informer [15] and Autoformer [16] repositories, respectively. To support reproducibility, the implementation of RWKV-CVM and the baselines, together with the exact chronological train/validation/test split indices and the three-seed evaluation scripts, are released at <https://github.com/xbtchen23/rwkv-cvm> (accessed on 1 March 2026).

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CVM	Cross-Variate Mixing
RWKV	Receptance Weighted Key Value
MTS	Multivariate Time Series
CI	Channel Independent
CD	Channel Dependent
STLF	Short-Term Load Forecasting
LTLF	Long-Term Load Forecasting
MSE	Mean Squared Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
WKV	Weighted Key Value
ADF	Augmented Dickey–Fuller (test)
CI95	95% Confidence Interval
DM	Diebold–Mariano (test)

Appendix A. Additional Prediction Visualizations

This appendix provides per-zone day-ahead ($H = 144$, 24 h) forecasts of RWKV-CVM, complementing the all-zone visualizations in Section 5.7. For each zone, windows at the 10th, 50th and 90th error percentiles are shown so that both typical and high-error (extreme-peak) behavior can be inspected. The model tracks the diurnal cycle well in the typical windows, while the 90th-percentile windows expose the larger deviations that occur around sharp evening peaks.

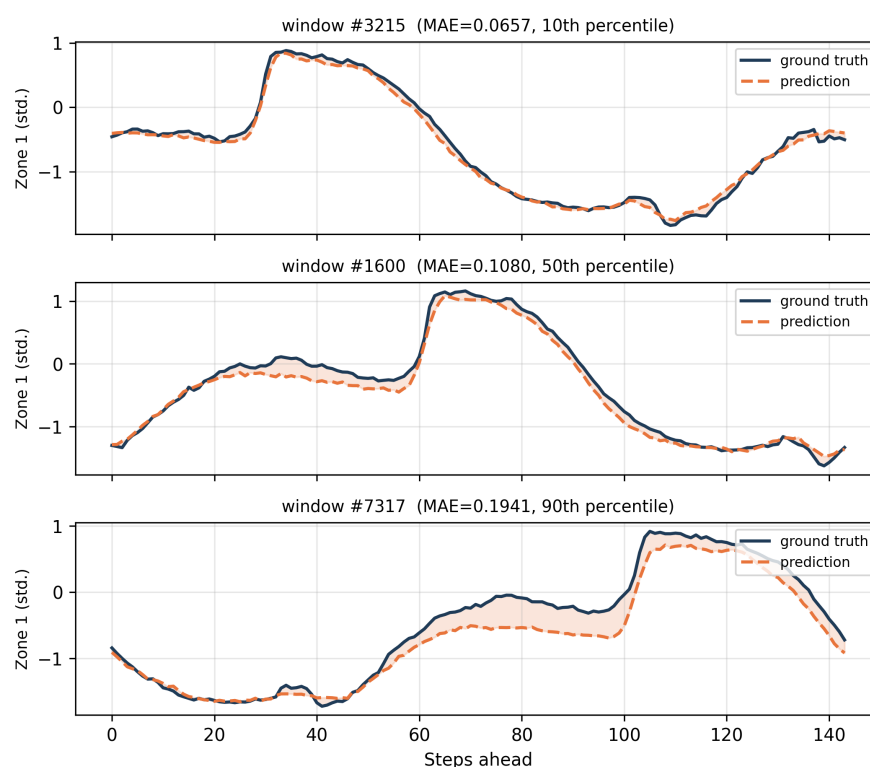
Tetouan all-zones (joint), RWKV-CVM(corr): Zone 1 forecast at H=144 (dayahead_24h)

Figure A1. Zone 1 day-ahead ($H = 144$, 24 h) forecasts of RWKV-CVM at the 10th, 50th and 90th error-percentile windows (standardized units).

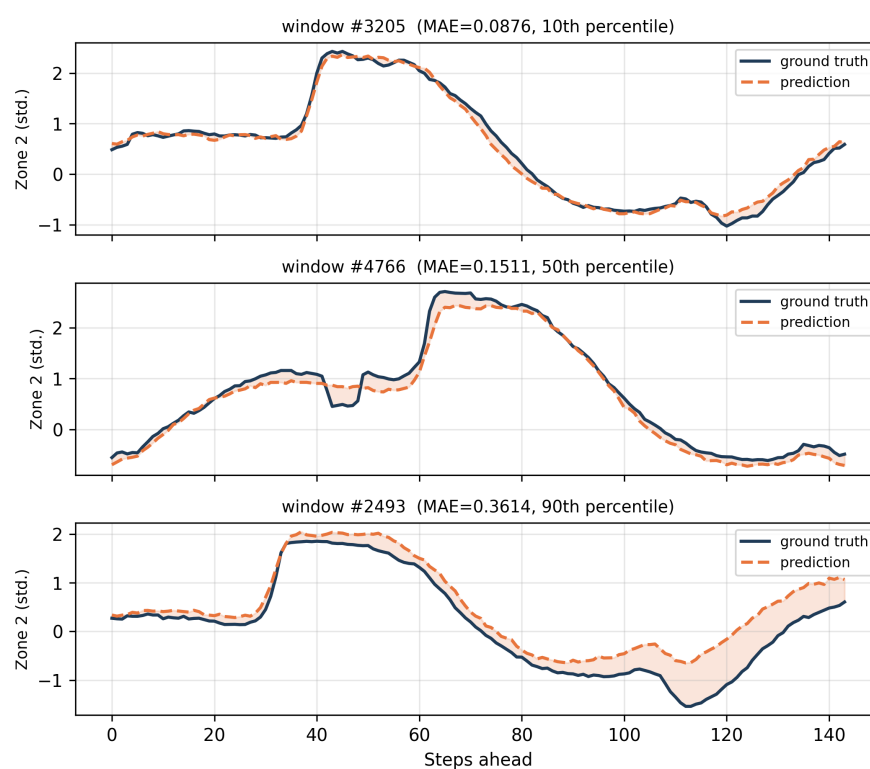
Tetouan all-zones (joint), RWKV-CVM(corr): Zone 2 forecast at H=144 (dayahead_24h)

Figure A2. Zone 2 day-ahead ($H = 144$, 24 h) forecasts of RWKV-CVM at the 10th, 50th and 90th error-percentile windows (standardized units).

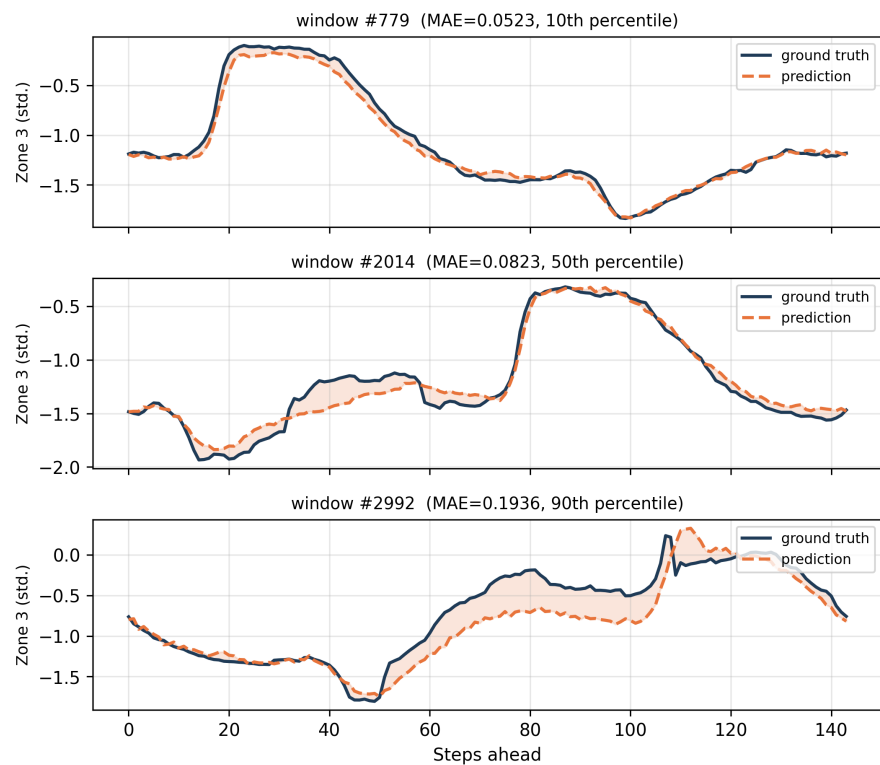
Tetouan all-zones (joint), RWKV-CVM(corr): Zone 3 forecast at H=144 (dayahead_24h)

Figure A3. Zone 3 day-ahead ($H = 144$, 24 h) forecasts of RWKV-CVM at the 10th, 50th and 90th error-percentile windows (standardized units). Zone 3 is the most autonomous zone (Section 5.4).

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